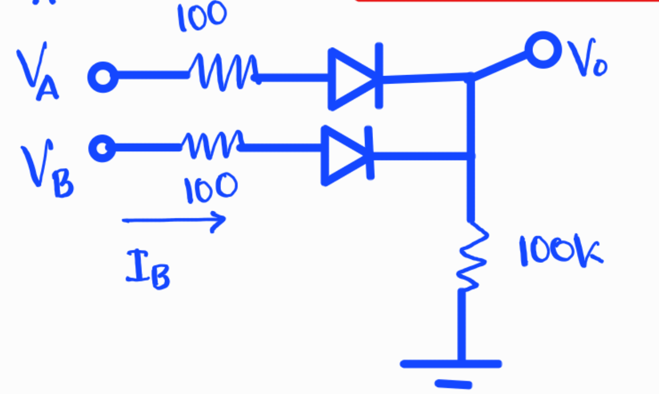
Set-A Q_1

ⓐ We choose $\boxed{R_1 = 100\Omega \quad R_2 = 100k\Omega}$



Case 1 Both input low

$$\boxed{V_o = 0V}$$

Case 2 Single input high

$V_A = 5V, V_B = 0V$

at node V_o, $\dfrac{V_o}{100k} = \dfrac{5 - V_o - 0.7}{100}$

$$\Rightarrow \boxed{V_o = 4.2957V}$$

Case 3 Both inputs high

at node V_o, $\dfrac{V_o}{100k} = 2 \times \dfrac{5 - V_o - 0.7}{100}$

$$\Rightarrow \boxed{V_o = 4.2978V}$$

ⓑ Maximum power consumption for case-3 (both input high)

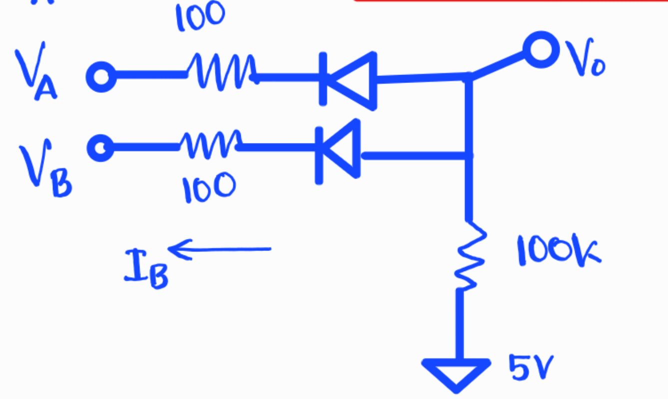
$$I_A = I_B = \frac{5 - 4.2978 - 0.7}{100}$$

$$= 2.2 \times 10^{-5} A$$

$$P = 2 \times I_A \times (5 - 0) = \boxed{2.2 \times 10^{-4} W}$$

Set-B Q_4

ⓐ We choose $\boxed{R_1 = 100\Omega \quad R_2 = 100k\Omega}$



Case 1 Both input high

$$\boxed{V_o = 5V}$$

Case 2 Single input high

$V_A = 5V, V_B = 0V$

at node V_o, $\dfrac{5 - V_o}{100k} = \dfrac{V_o - 0.7}{100}$

$$\Rightarrow \boxed{V_o = 0.7043 \ V}$$

Case 3 Both inputs low

at node V_o, $\dfrac{5 - V_o}{100k} = 2 \times \dfrac{V_o - 0.7}{100}$

$$\Rightarrow \boxed{V_o = 0.70214 \ V}$$

ⓑ Maximum power consumption for case-3 (both input low)

$$I_A = I_B = \frac{0.70214 - 0.7}{100}$$

$$= 2.14 \times 10^{-5} A$$

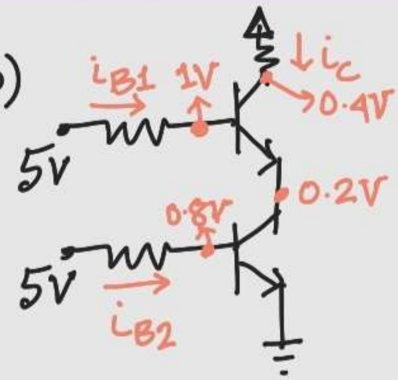
$$P = 2 \times I_A \times (5 - 0) = \boxed{2.14 \times 10^{-4} W}$$

Set-A (Q2)

(a) NAND gate.

V_A	V_B	T_1	T_2	v_o
0	0	cut	cut	5V
0	5	cut		5V
5	0		cut	5V
5	5	sat.	Sat.	0.4V

(b)

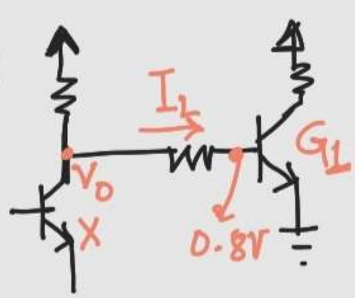


$$P = i_{B1}\times(5-0) + i_{B2}\times(5-0) + i_c\times(5-0)$$

$$220 = \frac{5-1}{0.4}\times 5 + \frac{5-0.8}{0.4}\times 5 + \frac{5-0.4}{R_c}\times 5$$

$$\Rightarrow R_c = \boxed{0.196\,k\Omega}$$

(c)



$$\frac{5-V_o}{R_c} = 5\times\frac{V_o-0.8}{5}$$

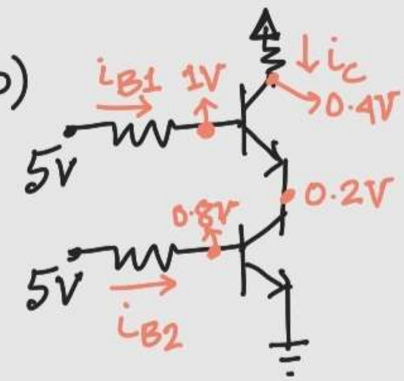
$$\Rightarrow v_o = \boxed{4.3117V}$$

Set-B (Q3)

(a) NAND gate.

V_A	V_B	T_1	T_2	v_o
0	0	cut	cut	5V
0	5	cut		5V
5	0		cut	5V
5	5	sat.	Sat.	0.4V

(b)

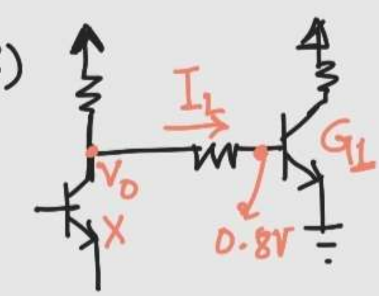


$$P = i_{B1}\times(5-0) + i_{B2}\times(5-0) + i_c\times(5-0)$$

$$175 = \frac{5-1}{0.3}\times 5 + \frac{5-0.8}{0.3}\times 5 + \frac{5-0.4}{R_c}\times 5$$

$$\Rightarrow R_c = \boxed{0.6\,k\Omega}$$

(c)



$$\frac{5-V_o}{R_c} = 6\times\frac{V_o-0.8}{6}$$

$$\Rightarrow v_o = \boxed{3.425\ V}$$

Set-A Q3

(a) $\boxed{0.2V}$ $V_{CE}(sat)$

(b)

$$\frac{V_P-0.7-0.2}{2}\times 3+\frac{V_P+12}{R_2+R_B}=\frac{V_{CC}-V_P}{R_1}$$

$$\Rightarrow V_P = 1.3016V$$

$$i_L = \frac{V_P-0.7-0.2}{2} = \boxed{0.2008\,mA}$$

$$I_1 = \frac{V_{CC}-V_P}{R_1} = \boxed{0.7132\,mA}$$

$$I_2 = \frac{V_P+12}{R_2+R_B} = \boxed{0.1108\,mA}$$

$$\boxed{I_B = I_C = 0}$$

(c) All inputs are low

$$V_B = V_P - I_2R_2$$
$$= -0.9144V$$

$V_{BE} < V_\gamma$; Cutoff

(If answered with all high)

$$I_1 = \frac{V_{CC}-0.8}{R_1+R_2} = 0.32\,mA$$

$$I_B = I_1 - \frac{0.8+12}{R_B} = 0.192\,mA$$

$$I_C = \frac{V_{CC}-0.2}{R_L} = 3.933\,mA$$

$$I_C/I_B = 20.49 < \beta_F \quad (Sat)$$

Set-B Q1

(a) $\boxed{0.2V}$ $V_{CE}(sat)$

(b)

$$\frac{V_P-0.7-0.2}{2}\times 3+\frac{V_P+12}{R_2+R_B}=\frac{V_{CC}-V_P}{R_1}$$

$$\Rightarrow V_P = 1.1841\ V$$

$$i_L = \frac{V_P-0.7-0.2}{2} = \boxed{0.1421\,mA}$$

$$I_1 = \frac{V_{CC}-V_P}{R_1} = \boxed{0.5408\,mA}$$

$$I_2 = \frac{V_P+12}{R_2+R_B} = \boxed{0.1146\,mA}$$

$$\boxed{I_B = I_C = 0}$$

(c) All inputs are low

$$V_B = V_P - I_2R_2$$
$$= -0.5349\ V$$

$V_{BE} < V_\gamma$; Cutoff

(If answered with all high)

$$I_1 = \frac{V_{CC}-0.8}{R_1+R_2} = 0.32\,mA$$

$$I_B = I_1 - \frac{0.8+12}{R_B} = 0.192\,mA$$

$$I_C = \frac{V_{CC}-0.2}{R_L} = 3.933\,mA$$

$$I_C/I_B = 20.49 < \beta_F \quad (Sat)$$

$V_{B0} = 0.8,\ V_0 = 0.1,$

$V_{C2} = 0.9,\ V_{B2} = 1.6,\ V_{B1} = 2.3$

Set-A-Q4

(a) $I_1 = \frac{6-2.3}{10k} = 0.37\text{ mA}$

$I_{EX} = I_{EY} = \beta_R I_1 = 0.037\text{mA}$

$I_{B2} = I_1 + 2I_{EX} = 0.444\text{ mA}$

$I_2 = \frac{5-0.9}{4k} = 1.275\text{mA}$

$I_{E2} = I_2 + I_{B2} = 1.72\text{ mA}$

$I_{B0} = I_{E2} - I_4 = 1.32\text{mA}$

$I_3 = \frac{6-0.1}{5k} = 1.18\text{mA}$

Individual Load current,

$I_L = I_1' = \frac{6-0.9}{10k} = 0.51\text{mA}$

$\therefore \frac{I_3 + NI_L}{I_{B0}} \leq \beta_F$

$\frac{1.18 + N(0.51)}{1.32} \leq 30$

$N = \lfloor 75.33 \rfloor = 75$

(b) $P = 6(I_1 + I_2 + I_3 + I_{EX} + I_{EY})$

$+ 10(0.1)(I_L)$

$= 17.904\text{ mW}$

(c) $\frac{1.18 + 70(0.51)}{1.32} < \beta_F$

$27.939 < \beta_F$

Set-B-Q2

(a) $I_1 = \frac{5-2.3}{5} = 0.54\text{mA}$

$I_{EX} = I_{EY} = \beta_R I_1 = 0.054\text{mA}$

$I_{B2} = I_1 + 2I_{EX} = 0.648\text{mA}$

$I_2 = \frac{5-0.9}{2k} = 2.05\text{mA}$

$I_{E2} = I_2 + I_{B2} = 2.698\text{mA}$

$I_{B0} = I_{E2} - I_4 = 2.298\text{mA}$

$I_3 = \frac{5-0.1}{4k} = 1.225\text{mA}$

Individual Load current,

$I_L = I_1' = \frac{5-0.9}{5k} = 0.82\text{mA}$

$\therefore \frac{I_3 + NI_L}{I_{B0}} \leq \beta_F$

$\frac{1.225 + N(0.82)}{2.298} \leq 40$

$N = \lfloor 110.603 \rfloor = 110$

(b) $P = 6(I_1 + I_2 + I_3 + I_{EX} + I_{EY})$

$+ 20(0.1)(I_L)$

$= 25.178\text{ mW}$

(c) $\frac{1.225 + 100(0.82)}{2.298} < \beta_F$

$36.216 < \beta_F$